Panel Data Models

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In *cross sectional analysis*:

- Observations on economic agents (e.g. individuals, households, firms, etc.) collected at a given point in time.
- Model given by

$$y_i = x_i'\beta + \varepsilon_i, \quad i = 1, \ldots, n.$$

where subscript i depends on individual i, and not on time t.

- We can follow the same random individual observations over time – known as *panel data* or *longitudinal data*.
- These units can be micro-level entities such as households, workers and firms, or macro-level entities such as cities, regions or countries.
- Typically, in micro panel data, we have a very large number of units relative to the number of time periods per unit; whereas in macro panel data we have fewer units and more time periods.
- In general, we have n units each of whom we follow for T_i consecutive time periods (for unit $i = 1, ..., n$).
- In this course we will focus on balanced panels in which T_i is the same for all i, and there are no missing observations.
- Individuals indexed by i, $i = 1, \ldots, n$; time indexed by t, $t = 1, \ldots, T$.
- **Important**: in this module we only look at panel data model where n is large and T is small. The size of the sample that matters here for the asymptotic theory to work is n (not T or $n \times T$).

Example of a data set: $n = 150$, $T = 2$.

A Two-Year Panel Data Set on City Crime Statistics

obsno	city	year	murders	population	unem	police
1	1	1986	5	350000	8.7	440
2	1	1990	8	359200	7.2	471
3	2	1986	2	64300	5.4	75
4	2	1990	1	65100	5.5	75
⋮	⋮	⋮	⋮	⋮	⋮	⋮
297	149	1986	10	260700	9.6	286
298	149	1990	6	245000	9.8	334
299	150	1986	25	543000	4.3	520
300	150	1990	32	546200	5.2	493

Advantages and Limitations of Panel Data

Advantages:

1. Controlling for individual unobserved heterogeneity
2. Identify individual-specific and time effects

Limitations of panel data:

1. Design and data collection problems
2. Usually short time-series dimension (this is why we focus on the case n large and T small).

Pooled Regression

The most restrictive model is a pooled model that specifies constant coefficients:

$$y_{it} = x'_{it}\beta + \varepsilon_{it}$$

- $\varepsilon_{it} \sim IID\left(0, \sigma^2_\varepsilon\right)$ meaning that:
    - $E[\varepsilon_{it}] = 0$,
    - $var[\varepsilon_{it}] = \sigma^2_\varepsilon$
    - the random variables (ε_{it}), $i = 1, ..., n$ and $j = 1, ..., T$ have the same distribution and are statistically independent over i and T.
- x_{it}: $k \times 1$ vector of explanatory variables (including constant).

OLS gives unbiased, consistent and efficient estimates of β if x_{it} is independent of ε_{it}.
This model is very simple and *doesn't allow* explicitly for individual heterogeneity.

Model:

$$y_{it} = \alpha_i + x'_{it}\beta + \varepsilon_{it} \tag{1}$$

- $\varepsilon_{it} \sim IID\left(0, \sigma_\varepsilon^2\right)$
- x_{it}: $k \times 1$ vector of explanatory variables.
- α_i : capture unobserved individual heterogeneity.
- In the modern formulation of this model [Wooldridge, (2002), Arellano (2003) and Cameron and Trivedi (2004)] the variables α_i are considered *random variables* and potentially correlated with x_{it} but independent of ε_{it}.

- Traditionally in the literature the variables α_i were treated as *fixed unknown parameters*, $i = 1, \ldots, n$; which would allow the intercept to vary over the individual units i.
- The estimator that we are going to propose for β is consistent and unbiased estimator for β even if we consider the variables α_i either as *fixed parameters* or as *random variables* and correlated with x_{it} but independent of ε_{it}.
- Henceforth, we will assume that the α_i are *random variables* and correlated with x_{it} but independent of ε_{it}.
- We require the regressors x_{it} to vary with t. Note that in some cases the regressors do not vary with t. **Examples:** race or gender of a person. The coefficients of these regressors are not identified.

- The model is

$$y_{it} = \alpha_i + x'_{it}\beta + \varepsilon_{it} \tag{2}$$

where

$$\varepsilon_{it} \sim IID\left(0, \sigma^2_\varepsilon\right), \quad \alpha_i \sim IID\left(0, \sigma^2_\alpha\right)$$

- x_{it}: $k \times 1$ vector of explanatory variables including a constant.
- α_i capture individual unobserved heterogeneity, but are *not correlated* with x_{it}: they are *random factors*. $\alpha_i + \varepsilon_{it}$ treated as an error consisting of two components: individual specific component α_i does not vary over t; remainder component ε_{it}, uncorrelated over t.
- This model permits identification of β but the assumption that α_i is uncorrelated with x_{it} is viewed as untenable in many microeconometrics applications.

- If you think there are *no omitted variables* – or if you believe that the omitted variables are uncorrelated with the explanatory variables that are in the model – then a *random effects* model is probably best.
- If you think there are *omitted variables*, and these variables are correlated with the variables in the model, then fixed effects models provide a mean for controlling for omitted variable bias. In this case, the omitted variables must have *time-invariant* values with time-invariant effects.
- If the key explanatory variable is *constant over time*, we cannot use fixed effects model and we must rely on the *random effects model*.
- If one uses random effects, as many time-constant controls as possible can be included.

Consider the *fixed effects model*:

- Define averages over t: $\bar{y}_i = T^{-1}\left(\sum_{t=1}^{T} y_{it}\right)$, $\bar{x}_i = T^{-1}\left(\sum_{t=1}^{T} x_{it}\right)$ and $\bar{\varepsilon}_i = T^{-1}\left(\sum_{t=1}^{T} \varepsilon_{it}\right)$.
- From (1):
$$\bar{y}_i = \alpha_i + \bar{x}_i'\beta + \bar{\varepsilon}_i. \tag{3}$$
- From (1) and (3):
$$y_{it} - \bar{y}_i = \left(x_{it} - \bar{x}_i\right)'\beta + \left(\varepsilon_{it} - \bar{\varepsilon}_i\right): \tag{4}$$
individual effects α_i are eliminated. Transformation that produces (4) known as **within transformation**.
- Let
$$\ddot{X}_i = \begin{bmatrix} (x_{i1} - \bar{x}_i)' \\ \vdots \\ (x_{iT} - \bar{x}_i)' \end{bmatrix}, \ddot{y}_i = \begin{bmatrix} y_{i1} - \bar{y}_i \\ \vdots \\ y_{iT} - \bar{y}_i \end{bmatrix}.$$

- Apply OLS to (4) to obtain the *within estimator* or *fixed effect estimator* given by:

$$
\begin{aligned}
\hat{\beta}_{within} &= \left[\sum_{i=1}^{n}\sum_{t=1}^{T}\left(x_{it}-\bar{x}_i\right)\left(x_{it}-\bar{x}_i\right)'\right]^{-1} \\
&\quad \times \left[\sum_{i=1}^{n}\sum_{t=1}^{T}(x_{it}-\bar{x}_i)(y_{it}-\bar{y}_i)\right] \\
&= [\sum_{i=1}^{n}\ddot{X}_i'\ddot{X}_i]^{-1}[\sum_{i=1}^{n}\ddot{X}_i\ddot{y}_i] \\
&= [W_{xx}]^{-1}W_{xy},
\end{aligned}
$$

where

$$
\begin{aligned}
W_{xx} &= \sum_{i=1}^{n}\sum_{t=1}^{T}\left(x_{it}-\bar{x}_i\right)\left(x_{it}-\bar{x}_i\right)', \\
W_{xy} &= \sum_{i=1}^{n}\sum_{t=1}^{T}(x_{it}-\bar{x}_i)(y_{it}-\bar{y}_i).
\end{aligned}
$$

- The term "*within*" comes from OLS using the time variation in y and x within each cross-sectional unit.

- The *least squares dummy variable* estimator is obtained assuming that the α_i are parameters to be estimated.
- Recall model:

$$y_{it} = \alpha_i + x_{it}'\beta + \varepsilon_{it}, \tag{5}$$

- Write model as:

$$y_{it} = \sum_{j=1}^{n} \alpha_j d_{ij} + x_{it}'\beta + \varepsilon_{it} \tag{6}$$

  where $d_{ij} = 1$ if $i = j$, and 0 otherwise.
- The parameters $\{\alpha_1, \ldots, \alpha_n, \beta\}$ can be estimated by OLS.
- This estimator of β referred to as *"least squares dummy variable" (LSDV) estimator*.
- It turns out that the estimator obtained for β using this approach is $\hat{\beta}_{within}$.

- If ε_{it} is independent of x_{is} for all s, t, then $\hat{\beta}_{within}$ is an unbiased estimator of β.
- As $n \to \infty$, consistency of $\hat{\beta}_{within}$ requires

$$E[(x_{it} - \overline{x}_i)\,\varepsilon_{it}] = 0. \tag{7}$$

Sufficient condition for (7) to hold

$$E[x_{it}\varepsilon_{it}] = 0 \text{ and } E[\overline{x}_i\varepsilon_{it}] = 0. \tag{8}$$

Condition in (8) implied by:

$$E[x_{is}\varepsilon_{it}] = 0 \text{ for all } s, t, \tag{9}$$

in which case we call x_{is} *strictly exogenous*. Condition in (9) satisfied if ε_{it} is independent of x_{is} for all s, t.

- If we consider the $\alpha_i's$ as parameters the estimator is given by estimation of the α_i:

$$\hat{\alpha}_i = \bar{y}_i - \bar{x}_i' \hat{\beta}_{within}, \quad i = 1, \ldots, n.$$

- Consistency of $\hat{\alpha}_i$ requires $T \to \infty$. As $n \to \infty$ (and T fixed), $\hat{\alpha}_i$ does not converge since the number of parameters α_i increases. Therefore, for large n and small T, $\hat{\alpha}_i$ is not consistent.

- *strictly exogenous* variables not allowed to depend upon current, future and past values of the error term. Strict exogeneity excludes inclusion of past values of y_{it} in x_{it}.
- If ε_{it} is homoskedastic with variance σ_ε^2,

$$\sqrt{n}(\hat{\beta}_{within} - \beta_0) \xrightarrow{d} \mathcal{N}(0, avar(\hat{\beta}_{within}))$$

  where

$$avar(\hat{\beta}_{within}) = \sigma_\varepsilon^2 E\left[\sum_{t=1}^{T} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'\right]^{-1}$$

  and $\hat{\beta}_{within}$ is *asymptotically efficient*.
- σ_ε^2 can be consistently estimated by

$$s_{within}^2 = (nT - n - k)^{-1} \sum_{i=1}^{n} \sum_{t=1}^{T} [(y_{it} - \bar{y}_i) - (x_{it} - \bar{x}_i)' \hat{\beta}_{within}]^2$$

- Hence the estimator for $avar(\hat{\beta}_{within})$ is

$$\widehat{avar}(\hat{\beta}_{within}) = s_{within}^2 \left[\frac{1}{n} \sum_{i=1}^{n} \sum_{t=1}^{T} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'\right]^{-1}$$

- The asymptotic distribution under *heteroskedasticity and serial correlation* is

$$\sqrt{n}(\hat{\beta}_{within} - \beta_0) \xrightarrow{d} \mathcal{N}(0, avar(\hat{\beta}_{within})),$$
$$avar(\hat{\beta}_{within}) = E[\ddot{X}_i'\ddot{X}_i]^{-1}E(\ddot{X}_i'\varepsilon\varepsilon_i'\ddot{X}_i)E[\ddot{X}_i'\ddot{X}_i]^{-1}$$

- An estimator robust to heteroskedasticity and serial correlation for small T/n is given by (Arellano, 1987)

$$\widehat{avar}(\hat{\beta}_{within}) = \left(\frac{1}{n}\sum_{i=1}^{n}\ddot{X}_i'\ddot{X}_i\right)^{-1}\frac{1}{n}\sum_{i=1}^{n}(\ddot{X}_i'\hat{u}_i\hat{u}_i'\ddot{X}_i)\left(\frac{1}{n}\sum_{i=1}^{n}\ddot{X}_i'\ddot{X}_i\right)^{-1}$$

where $\hat{u}_i = \ddot{y}_i - \ddot{X}_i\hat{\beta}_{within}$ are the fixed effect residuals for group i.

Estimation of the Fixed Effects Model (cont'd)

First Differences

- Another approach is to use first differences.
- Write the model for two years as

$$\begin{aligned} y_{it} &= \alpha_i + x_{it}'\beta + \varepsilon_{it}, \\ y_{i,t-1} &= \alpha_i + x_{i,t-1}'\beta + \varepsilon_{it-1},\ t = 2, ..., T \end{aligned}$$

- We can subtract one period from the other, to obtain

$$\Delta y_{i,t} = \beta' \Delta x_{i,t} + \Delta \varepsilon_i. \tag{10}$$

where $\Delta a_{i,t} = a_{it} - a_{it-1}$.
- Thus, with panel data, we can difference-out the unobserved fixed effect.
- Equation 10 is called the *first-differenced equation*, and the resulting OLS estimator of β is called the *first-differenced estimator*.

$$\begin{aligned} \hat{\beta}_{FD} &= \left[\sum_{i=1}^{n} \sum_{t=2}^{T} (\Delta x_{it}) (\Delta x_{it})'\right]^{-1} \\ &\quad \times \left[\sum_{i=1}^{n} \sum_{t=2}^{T} (\Delta x_{it}) (\Delta y_{it})\right] \end{aligned}$$

Estimation of the Fixed Effects Model (cont'd)

Fixed Effects of First Differencing?

- When we have only 2 time periods, $T = 2$, the *within* and *FD* estimates, as well as all test statistics, are *identical*, as long as we estimate the same model in each case (pay attention to the dummy variables!). If $T = 2$, *FD* is straightforward to implement and easy to compute heteroskedasticity-robust statistics, because *FD* is just a cross section.
- When $T \geq 3$, the *FD* and *within* estimators are not the same. Since both are unbiased and consistent, for large n and small T we need to examine their relative efficiency, which is determined by heteroskedasticity and serial correlation in the idiosyncratic errors, ε_{it}.
  - When ε_{it} are serially uncorrelated and homoskedastic, *within* is more efficient than *FD*.
  - When ε_{it} are substantially autocorrelated, in particular if ε_{it} follows a random walk (more on this in a future lecture), then $\Delta\varepsilon_{it}$ is serially uncorrelated, and *FD* is better.
  - When ε_{it} exhibit some serial correlation, but not as much as random walk, we cannot easily compare the efficiency of the *within* and *FD* estimators.

Estimation in the Random Effects (RE) Model

Consider the *random effects model*:

- Treats individual-specific effects α_i as random factors, independently and identically distributed over i (i.e. over individuals).
- Model:
$$y_{it} = x'_{it}\beta + \alpha_i + \varepsilon_{it} \tag{11}$$
$$\varepsilon_{it} \sim IID(0, \sigma^2_\varepsilon), \quad \alpha_i \sim IID(0, \sigma^2_\alpha)$$
- Term $\omega_{it} = \alpha_i + \varepsilon_{it}$ treated as an error term with two components:
  - *individual specific* α_i: does not vary over t;
  - *remainder component* ε_{it}: assumed to be uncorrelated over t
- Correlation of term ω_{it} over t due to α_i.
- Possible estimator: *Pooled OLS*. This estimator is *consistent*, but inefficient because it ignores the correlation among the ω_{it}. Notice also that valid inference with pooled OLS requires a *robust covariance estimator*.

Estimation of the Random Effects Model (cont'd)

- α_i and ε_{it} assumed to be *mutually independent*, and independent of x_{js} (for all j and s).
- For individual i, define $\iota_T = (1, \ldots, 1)'$ of dimension $T \times 1$ and $\varepsilon_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iT})'$, and denote by I_T the $T \times T$ identity matrix.
- Let $X_i = \begin{bmatrix} x'_{i1} \\ \vdots \\ x'_{iT} \end{bmatrix}$ and $y_i = \begin{bmatrix} y_{i1} \\ \vdots \\ y_{iT} \end{bmatrix}$ and $\omega_i = \alpha_i \iota_T + \varepsilon_i$.
- Hence the model can be written as

$$y_i = X_i\beta + \omega_i,\ i = 1, ..., n$$

- For individual i, variance matrix of $\omega_i = \alpha_i \iota_T + \varepsilon_i$:

$$\begin{aligned} Var\left[\omega_i\right] &= \Omega = \sigma_\alpha^2 \iota_T \iota_T' + \sigma_\varepsilon^2 I_T \qquad (12) \\ &= \begin{bmatrix} \sigma_\alpha^2 + \sigma_\varepsilon^2 & \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\alpha^2 + \sigma_\varepsilon^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\alpha^2 + \sigma_\varepsilon^2 & \cdots & \sigma_\alpha^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\alpha^2 & \cdots & \sigma_\alpha^2 + \sigma_\varepsilon^2 \end{bmatrix} \end{aligned}$$

Random Effects Model (cont'd)

- Let
$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}, \omega = \begin{bmatrix} \omega_1 \\ \vdots \\ \omega_n \end{bmatrix}$$
- Then the model can be written as $Y = X\beta + \omega$, where $E(\omega) = 0$ and $Var(\omega) = \Psi = I_n \otimes \Omega$.

Estimation of the Random Effects Model (cont'd)

- **Remark:** Let A be a $m \times n$ matrix with elements a_{ij} and B a $p \times q$ matrix then the *Kronecker product* of A and B is defined as the $mp \times nq$ matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}$$

- Hence

$$\begin{aligned} Var(\omega) &= I_n \otimes \Omega \\ &= \begin{bmatrix} \Omega & 0 & \cdots & 0 & 0 \\ 0 & \Omega & 0 & 0 & 0 \\ \vdots & 0 & \ddots & 0 & \vdots \\ 0 & 0 & 0 & \Omega & 0 \\ 0 & 0 & \cdots & 0 & \Omega \end{bmatrix} \end{aligned}$$

- A more efficient estimator than OLS can be obtained: *Generalised Least Squares* (GLS).

- Consider general case first

$$y = X\beta + \varepsilon \tag{13}$$

$$E[\varepsilon|X] = 0, \; Var[\varepsilon|X] = \Psi$$

  where Ψ is a positive definite matrix: model in (13) does not satisfy the Gauss-Markov assumptions.
- Transform model in (13) into one that satisfies Gauss-Markov assumptions.

Ψ positive definite $\Rightarrow$ can find nonsingular P such that $\Psi^{-1} = P'P$, so that

$$\Psi = (P'P)^{-1} = P^{-1}(P')^{-1}$$

$$P\Psi P' = PP^{-1}(P')^{-1}P' = I$$

- Pre-multiply ε in (13) by matrix P:

$$\begin{aligned} E[P\varepsilon|X] &= PE[\varepsilon|X] = 0 \\ Var[P\varepsilon|X] &= PVar[\varepsilon|X]P' = P\Psi P' = I : \end{aligned}$$

  $\Rightarrow P\varepsilon$ satisfies the *Gauss-Markov assumptions*.

- Transform whole model using P:

$$Py = PX\beta + P\varepsilon$$

  Write as

$$y^* = X^*\beta + \varepsilon^* \qquad (14)$$

  where

$$y^* = Py, \quad X^* = PX, \quad \varepsilon^* = P\varepsilon,$$

  and ε^* satisfies Gauss-markov assumptions.

Transformed model in (14) can be estimated by OLS to give the *BLUE* (Best Linear Unbiased Estimator) for β. This estimator is known as *Generalized Least Squares (GLS) Estimator* $\hat{\beta}_{GLS}$:

$$\begin{aligned}
\hat{\beta}_{GLS} &= (X^{*\prime}X^{*})^{-1}X^{*\prime}y^{*} \\
&= \left[(PX)'(PX)\right]^{-1}(PX)'(Py) \\
&= \left[X'(P'P)X\right]^{-1}X'(P'P)y \\
&= \left(X'\Psi^{-1}X\right)^{-1}X'\Psi^{-1}y
\end{aligned}$$

In practice Ψ is typically unknown and has to estimated in a first step. This gives the *Estimated GLS (EGLS) Estimator* or *Feasible GLS (FGLS) Estimator*.

We need to find the expression for P in the *random effects model*.

One can show that $P = I_n \otimes \Sigma$ where $\Sigma = [I_T - \frac{\theta}{T}\iota_T\iota_T']/\sigma_\varepsilon$ where

$$\theta = 1 - \frac{\sigma_\varepsilon}{\sqrt{\sigma_\varepsilon^2 + T\sigma_\alpha^2}}.$$

Hence we have to run the regression of PY on PX which can be shown to yield

$$\hat{\beta}_{GLS} = \left(\sum_{i=1}^n X_i'\Omega^{-1}X_i\right)^{-1}\sum_{i=1}^n X_i'\Omega^{-1}Y_i$$

It turns out that this is equivalent to run the regression

$$y_{it} - \theta\bar{y}_i = (x_{it} - \theta\bar{x}_i)'\beta + (\eta_{it} - \theta\bar{\eta}_i) \tag{15}$$

leading to

$$\begin{aligned}\hat{\beta}_{GLS} &= \left[\sum_{i=1}^n\sum_{t=1}^T (x_{it} - \theta\bar{x}_i)(x_{it} - \theta\bar{x}_i)'\right]^{-1} \\ &\quad \times \left[\sum_{i=1}^n\sum_{t=1}^T (x_{it} - \theta\bar{x}_i)(y_{it} - \theta\bar{y}_i)\right]\end{aligned}$$

- An alternative estimator for β in the random effect model is the *between estimator*.
- This consists in running the regression

$$\bar{y}_i = \bar{x}_i'\beta + \alpha_i + \bar{\varepsilon}_i. \tag{16}$$

Define the estimator $\hat{\beta}_B$ as

$$\begin{aligned} \hat{\beta}_B &= \left[\textstyle\sum_{i=1}^{n} \bar{x}_i \bar{x}_i{}'\right]^{-1} \textstyle\sum_{i=1}^{n} \bar{x}_i \bar{y}_i \\ &= [B_{xx}]^{-1} B_{xy}, \end{aligned}$$

where

$$\begin{aligned} B_{xx} &= \textstyle\sum_{i=1}^{n} \bar{x}_i \bar{x}_i{}', \\ B_{xy} &= \textstyle\sum_{i=1}^{n} \bar{x}_i \bar{y}_i \end{aligned}$$

- The *between estimator*: uses differences *between* individuals to estimate the parameters.

- It can be shown that

$$\hat{\beta}_{GLS} = [W_{xx} + \lambda B_{xx}]^{-1}[W_{xx}\hat{\beta}_{within} + \lambda B_{xx}\hat{\beta}_B],$$

$$\lambda = \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2} \tag{17}$$

  Thus $\theta = 1 - \lambda^{1/2}$.
- If $\lambda = 0$ then $\hat{\beta}_{GLS}$ becomes equal to *fixed effect estimator* $\hat{\beta}_{within}$; from (17), $\lambda \to 0$ if $T \to \infty$, so that $\hat{\beta}_{GLS}$ and $\hat{\beta}_{within}$ are equivalent only for large T.
- If $\lambda = 1$ then $\sigma_\alpha^2 = 0$: model in (11) has no individual-specific effects α_i. GLS estimator $\hat{\beta}_{GLS}$ becomes *pooled OLS* estimator since matrix Ω becomes diagonal.

- Consider the *transformed model*

$$(y_{it} - \theta\overline{y}_i) = (x_{it} - \theta\overline{x}_i)'\beta + u_{it} \tag{18}$$

- $u_{it} = (\eta_{it} - \theta\bar{\eta}_i)$ in (18) are $u_{it} \sim$ *IID* with respect to i and t.
- $\hat{\beta}_{within}$ corresponds to $\theta = 1$; OLS estimator $\hat{\beta}$ corresponds to $\theta = 0$.
- GLS estimator obtained as OLS estimator from model in (18): requires estimating θ and therefore $\lambda \Rightarrow$ *FGLS estimator*.
- From (17), estimation of λ (or θ) requires estimating σ_ε^2 and σ_α^2.

- Estimator for σ_ε^2 from *within group residuals*:

$$\hat{\sigma}_\varepsilon^2 = \frac{1}{n(T-1)} \sum_{i=1}^{n} \sum_{t=1}^{T} \left[y_{it} - \bar{y}_i - (x_{it} - \bar{x}_i)' \hat{\beta}_{within} \right]^2$$

- The error variance for the *between regression* is given by

$$\sigma_B^2 = \sigma_\alpha^2 + \frac{1}{T}\sigma_\varepsilon^2 \quad (19)$$

and estimated as

$$\hat{\sigma}_B^2 = \frac{1}{n} \sum_{i=1}^{n} \left(\bar{y}_i - \bar{x}_i' \hat{\beta}_B \right)^2 .$$

- From (19), σ_α^2 estimated as

$$\hat{\sigma}_\alpha^2 = \hat{\sigma}_B^2 - \frac{1}{T}\hat{\sigma}_\varepsilon^2.$$

- From (18), FGLS estimator from

$$(y_{it} - \hat{\theta}\overline{y}_i) = (x_{it} - \hat{\theta}\overline{x}_i)'\beta + error$$

is the *Random Effects estimator* for β (i.e. $\hat{\beta}_{RE}$) where

$$\hat{\theta} = 1 - \sqrt{\hat{\lambda}}, \quad \hat{\lambda} = \frac{\hat{\sigma}_\varepsilon^2}{\hat{\sigma}_\varepsilon^2 + T\hat{\sigma}_\alpha^2}.$$

Properties of Estimators in the RE Model

- We are working with large n and small T panels
- **Question:** given random effects model, which estimation method should we use?
- Suppose we use a *Fixed Effect* model specification:

$$y_{it} = \alpha_i + x_{it}'\beta + \varepsilon_{it}.$$

  Use within transformation to eliminate α_i and estimate the equation by OLS: this gives the *"within" estimator* from

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

  With strict exogeneity of x_{it}, $\hat{\beta}_{within}$ is consistent for β but *not efficient*: it ignores structure of the covariance matrix Ω.
- The *first differenced estimator* is also consistent in this framework, but not efficient.

Properties of Estimators in the RE Model

- *Between Estimator* $\hat{\beta}_B$ also consistent for β, but *not efficient*: it does not take into account information from within dimension.
- Consistent *and* asymptotically efficient estimator under assumption of x_{it} *strictly exogenous* *and* uncorrelated with α_i is the *RE Estimator* $\hat{\beta}_{RE}$.

Hausman Test

- Choice of estimation method requires testing whether α_i are correlated with the regressors or not.
- Assume strict exogeneity: $E[\varepsilon_{it} x_{is}] = 0, \forall s, t$.
- *Within estimator* consistent for β irrespective of whether x_{it} and α_i are correlated or not.
- *RE estimator* consistent for β *and* efficient: requires x_{it} and α_i are uncorrelated.
- Null hypothesis H_0: x_{it} and α_i are uncorrelated. Alternative hypothesis H_1: x_{it} and α_i are correlated.
- Under the null hypothesis the random effects model is the true model.

Hausman Test (cont'd)

- Test based on comparison between $\hat{\beta}_{within}$ and $\hat{\beta}_{RE}$.
- $\hat{\beta}_{within}$ *consistent* under H_0 and H_1.
- $\hat{\beta}_{RE}$ *consistent and efficient* under H_0, but *inconsistent* under H_1.
- Test statistic based on difference $(\hat{\beta}_{within} - \hat{\beta}_{RE})$.
- Under H_0:

$$avar\left[\hat{\beta}_{within} - \hat{\beta}_{RE}\right] = avar[\hat{\beta}_{within}] - avar[\hat{\beta}_{RE}].$$

- *Hausman Test statistic*:

$$\xi_H = n(\hat{\beta}_{within} - \hat{\beta}_{RE})' \left\{\widehat{avar}[\hat{\beta}_{within}] - \widehat{avar}[\hat{\beta}_{RE}]\right\}^{-1} (\hat{\beta}_{within} - \hat{\beta}_{RE})$$

where $\widehat{avar}\left[\cdot\right]$ denotes estimators for $avar\left[\cdot\right]$.

Hausman Test (cont'd)

- Under H_0, $\text{plim}(\hat{\beta}_{within} - \hat{\beta}_{RE}) = 0$ (since both estimators are consistent).
- Under H_0, ξ_H asymptotically distributed as

$$\xi_H \xrightarrow{D} \chi^2(k),$$

where k is the dimension of β.

- *Hausman test* tests whether $\hat{\beta}_{FE}$ and $\hat{\beta}_{RE}$ are significantly different.
- **Practical problem:** matrix $\left\{\widehat{avar}[\hat{\beta}_{within}] - \widehat{avar}[\hat{\beta}_{RE}]\right\}$ may not be positive definite in finite samples, so that its inverse may not be computed. These estimators dependent on the estimator used for σ^2_ε. To fix this problem one should use the same estimator for σ^2_ε in $\widehat{avar}[\hat{\beta}_{within}]$ and $\widehat{avar}[\hat{\beta}_{RE}]$.